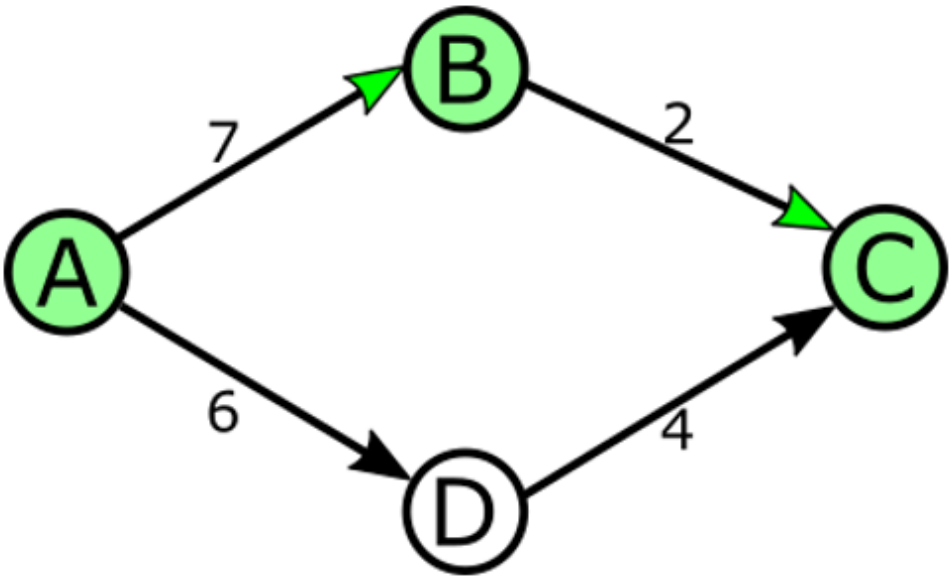


Figure 2.13 A directed graph. Highlighted, the shortest path between vertices `A` and `C`.

Prim's algorithm works exactly as Dijkstra's, except:

- Without keeping track of the distance from the source.
- Storing the edge that connected the front of the visited vertices to the next closest one.
- The vertex used as "source" for Prim's algorithm is going to be the root of the MST.

It should be no surprise that its running time is, similarly to Dijkstra's:

- `O(V2)` using arrays (sorted or unsorted) for the priority queue.
- `O(V*log(V) + E*log(V))` using binary or d-ary heap.
- `O(V*log(V) + E)` using Fibonacci heap.